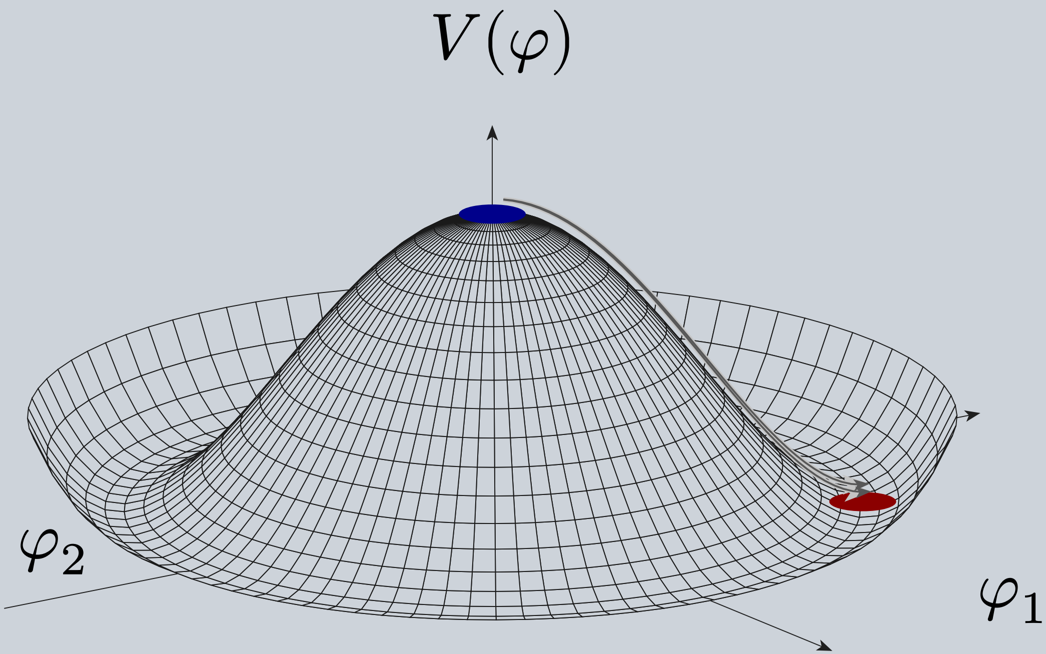
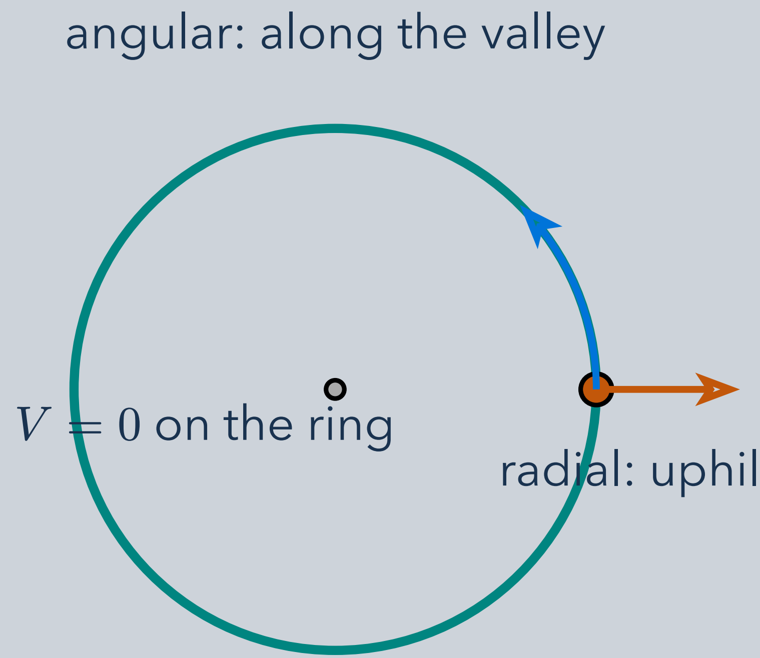
A symmetric energy landscape can have an entire family of equally good minima. Choosing one ground state need not preserve the symmetry of the equations.

### 1 A ring of minima




The peak at zero field is unstable; all points around the valley have the same potential energy. The selected red point breaks the rotational symmetry of this picture.

### 2 Radial and angular directions




For a global continuous symmetry, angular motion along the valley gives a massless Goldstone mode. Radial motion climbs the potential and costs energy.

**Higgs application:** in a gauge theory the would-be Goldstone modes supply longitudinal polarizations of massive gauge bosons; a radial Higgs excitation remains. The potential alone does not show the gauge-field dynamics.
$V(\varphi) = \lambda(|\varphi|^2 - v^2)^2$, with $\lambda > 0$.